Find the Error

Grade 12 · S.ID.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A data set has a mean of 10. An outlier of 100 is added to the high end. Does the mean increase?
Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Which measure of center is resistant to outliers: mean or median? Student's answer: mean

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

The data 1 through 9 gets an outlier of 100 added. What is the new range? Student's answer: 100

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 12 · S.ID.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A normal distribution has mean 100 and standard deviation 15. About 68% of values lie between which two numbers? Student's answer: 70 and 130

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

With mean 100 and standard deviation 15, what is the z-score of 130? Student's answer: 30

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

In a normal distribution, about what percent of values lie more than 1 standard deviation above the mean? Student's answer: 50%

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A survey of 100 students is taken to learn about the whole school. What is the whole school?
Student's answer: the sample

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

The mean of all voters' opinions is a parameter. Is it a parameter or a statistic? Student's answer: statistic

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A statistic is a number computed from which group? Student's answer: the population

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A model says a coin lands heads 10% of the time. In 50 flips, there are 20 heads. Is the model plausible? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A model says a die is fair. In 60 rolls, a 6 comes up 30 times. Is the model plausible? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A model says 50% of students like pizza. In a sample of 20, 11 like pizza. Is the model plausible?
Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A survey uses only people who choose to answer an online poll. Is this sample biased? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A survey asks: Don't you agree that homework is awful? Is the question biased? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A random sample of 100 is selected from a population. Is the sample mean an unbiased estimate of the population mean? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A random sample of 200 people has 120 who support a plan. Estimate the population proportion. Student's answer: 120

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A poll has a sample proportion of 60% and a margin of error of ±3%. What is the interval?
Student's answer: 3% to 63%

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A sample mean is 40 with a margin of error of ±2. What is the interval? Student's answer: 40 to 42

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Why are subjects randomly assigned to treatment and control groups? Student's answer: to make the groups different

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A randomized experiment finds a treatment causes improvement. Can we conclude the treatment caused the change? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Students who eat breakfast have higher grades in an observational study. Can we conclude breakfast causes higher grades? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

An observational study finds coffee drinkers have higher risk. A headline says coffee causes the risk. Is this conclusion supported? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A survey of 50 teachers in one school is reported as showing what all teachers think. Is this a valid generalization? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A random sample of 1,000 or a random sample of 10 gives an estimate. Which is more reliable?
Student's answer: 10

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

How many outcomes are in the sample space for flipping two coins? Student's answer: 2

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

How many outcomes are in the event 'at least one head' when flipping two coins? Student's answer: 1

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

On a standard die, what is the complement of the event 'rolling an even number'? Student's answer: {2, 4, 6}

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Are the events 'first flip is heads' and 'second flip is heads' independent? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

$P(A) = 0.5$, $P(B) = 0.4$, and $P(A \text{ and } B) = 0.2$. Are A and B independent? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

$P(A) = 0.3$, $P(B) = 0.6$, and $P(A \text{ and } B) = 0.3$. Are A and B independent? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

$P(A \text{ and } B) = 0.2$ and $P(B) = 0.5$. What is $P(A \mid B)$? Student's answer: 0.2

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A class has 18 girls and 12 boys. 6 girls wear glasses. What is P(glasses | girl)? Student's answer: 1/5

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

$P(A) = 0.4$ and $P(A \mid B) = 0.4$. Are A and B independent? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

In a survey of 100 people, 40 have trait A, 30 have trait B, and 15 have both. What is P(A or B)?
Student's answer: 0.70

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Using the same survey, what is P(A and B)? Student's answer: 0.40

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Using the same survey, what is P(B | A)? Student's answer: 0.15

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 12 · S.CP.5

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

$P(A) = 0.2$, $P(B) = 0.5$, and $P(A \text{ and } B) = 0.1$. Are A and B independent? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Events A and B are mutually exclusive, with $P(A) = 0.3$ and $P(B) = 0.4$. Are they independent?
Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

If $P(A \mid B) = P(A)$, are A and B independent? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

$P(A \text{ and } B) = 0.05$ and $P(B) = 0.25$. What is $P(A \mid B)$? Student's answer: 0.05

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A box has 50 items and 5 are defective. If the first item is defective and not replaced, what is P(second is defective)? Student's answer: 5/50

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

There are 40 seniors, and 15 of them are in honors. What is P(honors | senior)? Student's answer: 15/60

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

$P(A) = 0.5$, $P(B) = 0.4$, and $P(A \text{ and } B) = 0.2$. What is $P(A \text{ or } B)$? Student's answer: 0.9

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A and B are mutually exclusive, with $P(A) = 0.2$ and $P(B) = 0.3$. What is $P(A \text{ or } B)$? Student's answer: 0.06

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

$P(A) = 0.6$, $P(B) = 0.5$, and $P(A \text{ and } B) = 0.3$. What is $P(A \text{ or } B)$? Student's answer: 1.1

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A game pays $10 with probability 1/5 and costs $2 with probability 4/5. What is the expected value? Student's answer: 3.6

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is the expected value of rolling a fair six-sided die? Student's answer: 3

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A random variable X takes values 0, 1, and 2 with probabilities 0.5, 0.3, and 0.2. What is $E(X)$?
Student's answer: 1

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

X is 0 with probability 0.5 and 2 with probability 0.5. What is the variance of X? Student's answer: 2

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is the variance of rolling a fair six-sided die? Student's answer: 3.5

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A random variable has variance 9. What is its standard deviation? Student's answer: 9

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A ticket costs $2. It wins $6 with probability 1/4 and loses otherwise. Is this a fair game? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A game pays $10 with probability 0.1 and $0 otherwise. What is a fair price to play? Student's answer: $10

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A game pays $5 with probability 1/2 and $0 otherwise. What is a fair price to play? Student's answer: $5

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 12 · S.MD.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A product earns $100 with probability 0.6 and loses $50 with probability 0.4. What is the expected value? Student's answer: 60

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Option A has expected value 3 and option B has expected value 2.5. Which should be chosen to maximize expected value? Student's answer: B

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

An insurance premium is $200. The chance of a $5,000 loss is 0.02. Is the premium lower than the expected loss? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A player wins if a fair die shows an even number and loses otherwise. Is this game fair? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Four students use a fair spinner with 4 equal sections to choose one student for a prize. What is each student's chance? Student's answer: 1/3

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Two dice are rolled. A player wins if the sum is 7 or more, and the other player wins otherwise. Is this fair? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.